

Unit 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit develops the treatment of chemical bonding by introducing intermediate types of bonding and by exploring the nature and effects of intermolecular forces.

Study of the Periodic Table is extended to cover the chemistry of Groups 1, 2 and 7, where ideas about redox reactions are applied to the reactions of halogens and their compounds.

The study of energetics in chemistry is of theoretical and practical importance. In this unit, students learn how to define, measure and calculate enthalpy changes. They will see how a study of enthalpy changes helps chemists to understand chemical bonding.

The unit also develops an understanding – mostly at a qualitative level – of the ways in which chemists can control the rate, direction and extent of chemical change in reactions.

The organic chemistry in this unit covers halogenoalkanes and alcohols, and explores the mechanisms of selected reactions.

The study of spectroscopy gives further examples of the importance of accurate and sensitive methods of analysis, which can be applied to study chemical changes but also to detect drugs such as ethanol.

Chemistry in action

The use of models in chemistry is illustrated by the way in which the Maxwell-Boltzmann distribution and collision theory can account for the effects of temperature on the rates of chemical reactions.

The unit shows how chemists can study chemical changes at an atomic level and propose mechanisms to account for their observations.

The study of rates and equilibria shows the contribution that chemistry can make to a more sustainable economy by redeveloping manufacturing processes to make them more efficient, less hazardous and less polluting.

Practical skills

Again, students can begin their practical work for this unit with simple reactions, in polystyrene cups, to investigate energy changes in chemical reactions. This leads to the first core practical in this unit.

Inorganic chemistry and basic redox reactions can also be explored with simple test-tube reactions.

The techniques of volumetric analysis are introduced in this unit, with two core practicals to develop competence in this key skill.

The effect of temperature, concentration and surface area on the rate of a reaction can be explored through a variety of reactions, and also forms part of the first core practical in organic chemistry.

Further core practicals in organic chemistry are used to develop students' skills in using glassware and techniques such as reflux, use of a separating funnel and distillation.

The final core practical considers qualitative analysis for ions and organic functional groups.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit. This includes plotting and extrapolating graphs of temperature rise against time for displacement reactions, calculating enthalpy changes in J and kJ mol^{-1} , using algebra to solve Hess's Law problems, calculating enthalpy changes using bond enthalpies and evaluating experimental results in terms of measurement uncertainties and systematic errors in the context of measuring energy changes and in titrations. Also, calculating oxidation numbers within a complex system, balancing equations for redox reactions by combining ionic half-equations, calculating rates from reaction time, plotting graphs and having an appreciation of the graph for a Maxwell-Boltzmann distribution, and deriving an algebraic expression for the equilibrium constant. Further mathematical skills can be developed in analysing mass and infrared spectra. (Please see *Appendix 6: Mathematical skills and exemplifications* for more details.)

Assessment information

- First assessment: June 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 80 marks.
 - Students must answer all questions.
 - This paper has three sections:
 - Section A: multiple choice questions
 - Section B: mixture of short-open, open-response, calculations and extended-writing questions
 - Section C: contemporary context question.
 - This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
 - This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
 - Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
 - This paper may contain some synoptic questions which require knowledge and understanding from Unit 1.
 - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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Topic 6: Energetics

Students will be assessed on their ability to:

6.1	know that the enthalpy change, ΔH , is the heat energy change measured at constant pressure and that standard conditions are 100 kPa and a specified temperature, usually 298 K
6.2	know that, by convention, exothermic reactions have a negative enthalpy change and endothermic reactions have a positive enthalpy change
6.3	be able to construct and interpret enthalpy level diagrams, showing exothermic and endothermic enthalpy changes
6.4	know the definition of standard enthalpy change of: i reaction, $\Delta_r H$ ii formation, $\Delta_f H$ iii combustion, $\Delta_c H$ iv neutralisation, $\Delta_{\text{neut}} H$ v atomisation, $\Delta_{\text{at}} H$
6.5	be able to use experimental data to calculate: i energy transferred in a reaction recalling and using the expression: energy transferred (J) = mass (g) $\times$ specific heat capacity ($\text{J g}^{-1} \text{ }^\circ\text{C}^{-1}$) $\times$ temperature change ($^\circ\text{C}$) ii enthalpy change of the reaction in kJ mol^{-1} <i>This will be limited to experiments where substances are mixed in an insulated container and combustion experiments using a suitable calorimeter.</i>
6.6	know Hess's Law and be able to apply it to: i constructing enthalpy cycles ii calculating enthalpy changes of reaction using data provided, or data selected from a table or obtained from experiments
6.7	CORE PRACTICAL 2 Determination of the enthalpy change of a reaction using Hess's Law.
6.8	be able to evaluate the results obtained from experiments and comment on sources of error and uncertainty and any assumptions made in the experiments <i>Students will need to consider experiments where substances are mixed in an insulated container and combustion experiments using, for example, a spirit burner and be able to draw suitable graphs and use cooling curve corrections.</i>
6.9	understand the terms 'bond enthalpy' and 'mean bond enthalpy', and be able to use bond enthalpies to calculate enthalpy changes, understanding the limitations of this method
6.10	be able to calculate mean bond enthalpies from enthalpy changes of reaction

Further suggested practicals:

- i the enthalpy change for the decomposition of calcium carbonate using the enthalpy changes of reaction of calcium carbonate and calcium oxide with hydrochloric acid
- ii the enthalpy change of combustion of an alcohol
- iii the enthalpy change of the reaction between zinc and copper(II) sulfate solution
- iv the enthalpy of hydration of anhydrous copper(II) sulfate